Test Plan

TS-0078 — APP-002 — Performance Suite 78

1. Test Plan Overview

Suite ID	TS-0078
Suite Name	APP-002 — Performance Suite 78
Suite Type	Performance
Priority	High
Status	Draft
Application	APP-002 — MobileCommunication
Release	REL-007 — Release 5.3.0
Release Version	5.3.3
Release Date	2023-10-12
Created Date	2023-04-29

2. Scope & Objectives

This test plan covers performance testing for the MobileCommunication application. Suite covering Cache scenarios for APP-002.

3. Test Strategy

Testing will follow a risk-based approach with priority on High severity scenarios. Automated test cases will be executed first to detect regressions early, followed by manual exploratory testing for edge cases. Performance and security test cases will run in parallel in dedicated environments.

4. Requirements Under Test

REQ-0433	Reliability — Payment Processing	Reliability	Low	Approved
REQ-0069	UI Behavior — Remote Diagnostics	Reliability	Low	Approved
REQ-0390	Data Synchronization — Configuration Reload	Compatibility	Medium	Deprecated
REQ-0153	Network Behavior — Search Query Execution	Battery	Medium	Approved
REQ-0313	UI Behavior — Background Sync	Security	High	Deprecated

5. Test Cases Summary

TC-0061	Validate that session token refresh enforces acces...	Smoke	Manual	Low
TC-0291	Confirm that the system recovers gracefully after ...	System	Automated	Low
TC-0302	Ensure that user authentication does not exceed th...	Security	Manual	Critical
TC-0497	Validate that Bluetooth reconnection enforces acce...	Sanity	Semi-Automated	Medium
TC-0586	Validate that session token refresh enforces acces...	Functional	Automated	Low

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TC-0615	Ensure that NFC tap-to-pay does not exceed the p95...	Sanity	Semi-Automated	Low
TC-0776	Test that media streaming correctly handles invali...	Functional	Semi-Automated	Low
TC-0887	Confirm that device pairing triggers the expected ...	Security	Manual	Low

6. Test Environments

Primary	Mobile Lab — Android 14 — WiFi-6 (ENV-001)
Secondary	Staging — Android 13 — LTE (ENV-007)
Extended	Low Network Env — Android 14 — Low-BW Sim (ENV-009)
Performance	High Load Env — Ubuntu 22.04 — Ethernet (ENV-010)

7. Entry & Exit Criteria

Entry — Build	Smoke suite must pass 100% before full regression begins.
Entry — Env	All required environments must be Available status.
Entry — Defects	No Critical-severity open defects from the previous cycle.
Exit — Pass Rate	Overall pass rate >= 95%.
Exit — Critical	Zero open Critical-severity defects.
Exit — Coverage	All Priority=High test cases executed.

8. Risks

Network instability in low-BW env	Medium	Reserve stable WiFi fallback
Device unavailability	Low	Reserve backup devices 48h ahead
Build quality regression	Medium	Execute smoke before full regression
Flaky automation scripts	Low	Quarantine flaky TCs, re-run 3x

9. Responsible Engineers

ENG-050	Zach Singh	Security Tester	XR QA
ENG-033	Xavier Evans	Automation Engineer	Security QA
ENG-031	Laura Rossi	QA Engineer	Automation QA

10. Synthetic Data Notice

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